

Solar System

Hayrettin Yazıcı - 150240310

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1 Introduction

Our solar system includes the Sun, eight planets, five officially named dwarf planets, and hundreds of moons, and thousands of asteroids and comets. Our solar system is located in the Milky Way, a barred spiral galaxy with two major arms, and two minor arms. Our Sun is in a small, partial arm of the Milky Way called the Orion Arm, or Orion Spur, between the Sagittarius and Perseus arms. Our solar system orbits the center of the galaxy at about 515,000 mph (828,000 kph). It takes about 230 million years to complete one orbit around the galactic center.

2 Planets

The solar system has eight planets:

- Mercury
- Venus
- **Earth**
- Mars
- Jupiter
- Saturn
- Uranus
- Neptune

An example drawing of solar system can be seen in the Figure 1. 1

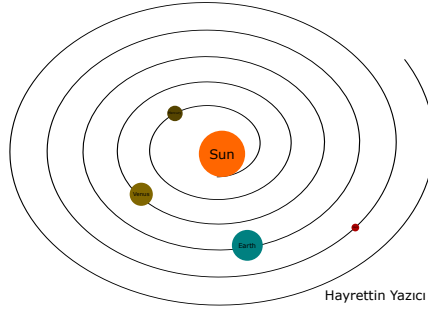


Figure 1: Solar System Drawing

2.1 Planet Sizes and Locations

Planet	Equatorial Diameter (km)	Order from the Sun
Mercury	4,880	1
Venus	12,104	2
Earth	12,756	3
Mars	6,792	4
Jupiter	142,984	5
Saturn	120,536	6
Uranus	51,118	7
Neptune	49,528	8

Table 1: Names, diameters, and orbital positions of planets

As shown in Table 1, the planets in our solar system vary greatly in size and distance from the Sun. For instance, **Jupiter**, with a diameter of 142,984 km, is the *largest*, while **Mercury**, the *smallest* planet, has a diameter of only 4,880 km. These variations highlight the diversity among the planets orbiting our Sun.